

# LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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## Abstract

Multicenter medical models are typically trained asynchronously across hospitals with heterogeneous acquisition devices, patient populations, and diagnostic-label coverage. We study single-shot asynchronous medical model merging, where each client uploads its locally trained model only once and the server constructs a global model without accessing raw medical images or conducting federated optimization rounds. We identify aggregation-induced prediction collapse as the central failure mode of general-purpose model-merging methods. Furthermore, we show that long-tailed medical priors can make majority-class collapse deceptively competitive in terms of raw accuracy, thereby masking this failure. To address this issue, we propose LAMP-Merge, a long-tail-aware medical prototype-merging framework that reconstructs a diagnostic prototype head from client-side class statistics and calibrates long-tail prevalence using a bounded prior bias. Across five medical datasets, four vision-model families, three client-population sizes, and three levels of non-IID skew, LAMP-Merge matches or surpasses the strongest non-LAMP baseline in 57 of 60 client-averaged settings, while improving balanced accuracy, macro-F1, and prediction-collapse diagnostic metrics.

Index Terms: medical model merging, asynchronous fusion, predictive collapse, long-tail calibration, privacy-preserving learning

## I Introduction

Medical artificial intelligence increasingly requires **multi-center collaboration** because no single hospital can cover the full disease spectrum, rare cases, and imaging conditions (Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021). Yet privacy requirements, device heterogeneity, and institutional workflows make **centralizing raw images impractical** (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016). A realistic alternative is for hospitals to train locally and share limited post-training information.

This setting requires **single-shot asynchronous model merging**. Hospitals may complete training at different times and cannot necessarily participate in repeated global updates, as assumed by conventional federated and decentralized training protocols (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Warnat-Herresthal et al. 2021). The server must therefore construct one deployable diagnostic

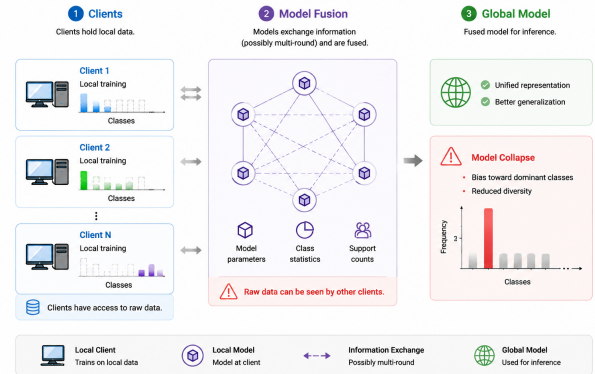


Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model’s predictions to concentrate on a small number of dominant classes.

model from **exactly one upload per client**, without accessing raw medical images.

Medical imaging makes the problem especially difficult because hospitals differ not only in device domain but also in **diagnostic-class availability** (Guan and Liu 2022; Zech et al. 2018; Li et al. 2021). Local models are partial experts: a client may provide strong evidence for common diagnoses while lacking rare ones. Moreover, medical labels are often **long-tailed**, so a useful merge must preserve distributed class-specific evidence while respecting genuine prevalence (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

We propose **LAMP-Merge**, a long-tail-aware medical prototype-merging framework. It recasts global parameter averaging as **class-level diagnostic-evidence reconstruction**: each client uploads class prototypes and counts computed with a shared reference extractor, and the server constructs a global prototype classifier with bounded prevalence calibration. The procedure requires one upload and never exposes raw images, per-sample features, logits, or predictions.

Our main contributions are as follows:

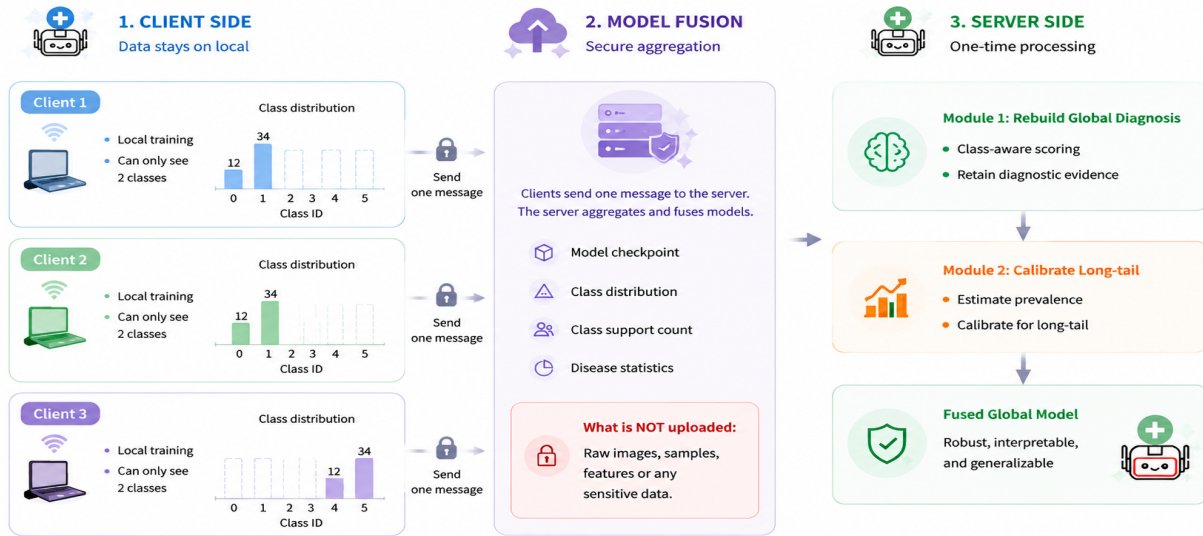


Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through diagnostic prototype reconstruction and long-tail prevalence calibration, without accessing raw data or requiring multi-round federated optimization.

- We formalize **single-shot asynchronous medical image model merging**, where independently trained clients provide one upload for global model construction.
- We identify **heterogeneous diagnostic-class availability and long-tailed disease prevalence** as the defining structures of this setting.
- We propose **LAMP-Merge**, which reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics to mitigate prediction collapse without exposing raw images.

## II Related Works

### Model Merging

Model-merging methods commonly integrate checkpoints in parameter, task-vector, or geometric spaces. Weight averaging and Model Soups interpolate model parameters (Izmailov et al. 2018; Wortsman et al. 2022); permutation-aware matching aligns hidden-unit symmetries before averaging (Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023); ZipIt performs feature-level zipping across models (Stoica et al. 2024); Task Arithmetic, TIES-Merging, and DARE operate on task deltas and sign conflicts (Ilharco et al. 2023; Yadav et al. 2023; Yu et al. 2024); Fisher merging, RegMean, Model Stock, Bread-crumbs, Iso-C, FreeMerge, and RobustMerge further introduce curvature, mean-statistic, subspace, or robustness assumptions (Matena and Raffel 2022; Jin et al. 2025; Jang, Yun, and Han 2025; Davari and Belilovsky 2024; Marczak et al. 2025; Zheng and Wang 2025; Zeng et al. 2025).

However, these variables are still global parameters, task vectors, or update directions rather than class-level diagnos-

tic evidence. **They do not ask whether a client has reliable supervision for a specific medical class.** Under incomplete local class coverage and long-tailed prevalence, reliable directions from clients that observed a class can be averaged with absent or conflicting directions from clients that did not, leading to aggregation-induced prediction collapse (Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

### Medical Model Merging

Medical fusion studies often exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN, Raindrop, and MedFuse show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations (Shukla and Marlin 2021; Zhang et al. 2022; Hayat, Geras, and Shamout 2023).

However, these methods usually fuse patient-level inputs or clinical representations rather than independently trained image-classifier checkpoints, and they often require patient samples, timestamps, multimodal records, electronic health records, or server-side validation feedback, which conflicts with common medical privacy constraints (Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). MedMerge is a relevant training-free medical knowledge-transfer framework (Wei et al. 2026); however, it targets cross-modal adaptation for medical vision-language models rather than asynchronous checkpoint merging for the same classification task. **Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints.** LAMP-Merge keeps the medical structure uploadable by using only class-level prototypes and prevalence counts computed locally by clients.

## Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg and its medical deployments rely on repeated server-client updates (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021); Swarm Learning and Gossip Learning reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states (Warnat-Herresthal et al. 2021; Heged\Hus, Danner, and Jelasity 2019).

However, our problem is single-shot asynchronous model merging after local training has finished: hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. **The limitation is therefore communication feasibility rather than only privacy.** LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

## III Method

### Motivation and Overall Idea

Under partial local class coverage, parameter-level merging can turn several non-degenerate local predictors into a globally collapsed predictor. Because medical prevalence is often long-tailed, such collapse can still appear competitive in raw accuracy. A useful merger must therefore recover rare-class discrimination without discarding genuine prevalence.

LAMP-Merge addresses this tension with two components. Diagnostic prototype reconstruction aggregates only class-supported evidence in a shared reference space to recover global multiclass directions. Long-tail prevalence calibration then applies a bounded prior bias estimated from client-side counts, preserving clinically meaningful majority prevalence without replacing those directions. The resulting one-shot protocol requires neither additional federated training nor raw-data exposure.

### Overview

The core framework of LAMP-Merge is illustrated in figure 2. Its central idea is to recast medical model merging from global parameter averaging into the reconstruction of class-conditional diagnostic evidence, supplemented by class-prior information.

Notably, LAMP-Merge neither requires the server to access clients' raw images nor relies on a server-side validation set, per-sample features, or per-sample predictions. Clients upload only class-level statistics; thus, the framework preserves a single-shot asynchronous communication boundary, distinguishing it from federated learning procedures that require repeated model broadcasts and local retraining.

The framework comprises two key components: a class-level diagnostic prototype reconstruction module and a long-tail prevalence calibration module. The prototype reconstruction module restores multiclass discriminative directions across the complete global diagnostic space, after

which the calibration module retains appropriate class-prior information.

### Diagnostic Prototype Reconstruction Module

Assume  $K$  medical clients. Client  $i$  owns a private labeled dataset  $D_i = \{(x, y)\}$  drawn from a class-skewed medical distribution over  $C$  diagnosis classes. Let  $D_{i,c}$  denote the subset of client  $i$  belonging to diagnosis class  $c$ . In medical partitions,  $D_{i,c}$  can be empty for many classes. Thus, a client can be reliable for a subset of diagnoses but unreliable for unseen classes. A class-agnostic parameter merge has no explicit mechanism to preserve this local reliability structure.

Therefore, LAMP-Merge first reconstructs a class-wise diagnostic head in the shared reference feature space. Let  $\phi_0$  be the frozen reference backbone instantiated from the shared model configuration before local training, and let  $T(\cdot)$  be the evaluation transform. For each diagnosis class  $c$ , client  $i$  computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y) \in D_i} \mathbf{1}[y = c] \phi_0(T(x)), \quad (1)$$

where  $n_{i,c} = |D_{i,c}|$  is the class support count. If  $n_{i,c} = 0$ , the client uploads zero support for that class and contributes no prototype evidence.

The server converts support counts into evidence weights:

$$e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0], \quad (2)$$

where  $\gamma = 0.45$  is the default evidence exponent. The normalized reliability of client  $i$  for class  $c$  is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^K e_{j,c}}. \quad (3)$$

The global diagnosis prototype is then

$$p_c = \sum_{i=1}^K \alpha_{i,c} \mu_{i,c}. \quad (4)$$

Finally, LAMP-Merge synthesizes a cosine prototype classifier:

$$w_c = s \frac{p_c}{\|p_c\|_2}, \quad (5)$$

where  $s = 20$  is the default head scale. This module addresses a key limitation of existing methods: it preserves the classes for which each client has supporting evidence and reconstructs a global multiclass diagnostic head.

### Long-tail Prevalence Calibration

After obtaining this newly reconstructed diagnostic head, each client additionally uploads class-prevalence counts  $m_{i,c}$  to account for the long-tailed prevalence inherent in real-world medical tasks. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^K m_{i,c}}{\sum_{k=1}^C \sum_{i=1}^K m_{i,k}}. \quad (6)$$

We measure dominant-class imbalance by

$$r = C \max_c \pi_c. \quad (7)$$

The module is activated only when  $r > \tau$ , where  $\tau = 2.5$  by default. When activated, it adds a centered log-prior bias

$$b_c = \lambda \left( \log \pi_c - \frac{1}{C} \sum_{k=1}^C \log \pi_k \right), \quad (8)$$

with default maximum calibration strength  $\lambda = 5.0$ . The final score is

$$\text{score}_c(x) = w_c^\top \phi_0(T(x)) + b_c. \quad (9)$$

The bias is centered, bounded, and attached to the prototype classifier; it calibrates prevalence without replacing the multi-class diagnostic directions reconstructed by the Diagnostic Prototype Reconstruction Module.

## Experiments

### Experimental Settings

**1) Evaluation Datasets:** We evaluate the merged models on five representative MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification (detailed dataset descriptions are provided in the Appendix).

**2) Baselines:** We compare LAMP-Merge with twelve model-merging baselines covering the main technical routes of current model merging: `avg` (Wortsman et al. 2022), `ties` (Yadav et al. 2023), `dare_linear` (Yu et al. 2024), `dare_ties` (Yu et al. 2024), `regmean` (Jin et al. 2025), `fisher` (Matena and Raffel 2022), `breadcrumbs` (Davari and Belilovsky 2024), `model_stock` (Jang, Yun, and Han 2025), `from` (Ilharco et al. 2023), `iso_c` (Marczak et al. 2025), `free_merge` (Zheng and Wang 2025), and `robustmerge` (Zeng et al. 2025). These baselines include weight averaging, task-vector pruning, sign-conflict resolution, curvature or mean-statistic correction, stock-style model interpolation, isotropic subspace merging, Fourier-domain fusion, and direction-robust merging.

**3) Merging Settings:** Experiments follow a single-shot asynchronous model-merging protocol. Each client first independently trains a local model on its own medical imaging data, then computes the class-level statistics required by LAMP-Merge locally, including class-support counts, reference-space class prototypes, and class-prevalence counts. These statistics are uploaded once together with an interface to the local checkpoint. The server subsequently performs a single model-merging step using only the uploaded checkpoints and class-level statistics, and evaluates the resulting merged model on a global test set.

We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny (He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with  $K \in \{3, 5, 7\}$  clients and Dirichlet skew parameter  $\beta \in \{0, 0.01, 0.1\}$ , using seed 42 (Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and  $K$ , and then averages over the three  $\beta$  values.

**4) Implementation Environment:** All experiments were run on a server with two Intel(R) Xeon(R) Silver 4210 CPUs, four NVIDIA GeForce RTX 2080 Ti GPUs, and the conda environment MM with Python 3.10.20, PyTorch 2.6.0+cu118, and CUDA runtime 11.8.

### Comparison With the State of the Arts

The main comparison uses client-average ACC: for each fixed dataset, model, and client number, we first average over the three  $\beta$  skew levels and then compare merging methods. A cell is counted as successful when LAMP-Merge is greater than or equal to the strongest non-LAMP baseline in that cell.

Tables 1–4 report the full client-average results for all four backbones. Across the 60 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in **57 cells (95.0%)**.

Table 1: Client-average test accuracy on ResNet. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over  $K = 3, 5, 7$  within the corresponding dataset.

| Method                                  | Blood         |               |               |               | Derma         |               |               |               | Organ-C       |               |               |               | Organ-S       |               |               |               | Ultrasound    |               |               |               |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                                         | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           |
| avg(Wortsman et al. 2022)               | 0.3345        | 0.2467        | 0.2822        | 0.2878        | 0.6291        | 0.4700        | 0.6244        | 0.5745        | 0.2261        | 0.1772        | 0.1396        | 0.1810        | 0.2864        | 0.1794        | 0.2475        | 0.2378        | 0.2435        | 0.1581        | 0.1542        | 0.1853        |
| ties(Yadav et al. 2023)                 | 0.2601        | 0.1872        | 0.1787        | 0.2087        | 0.5995        | 0.4841        | 0.6484        | 0.5773        | 0.2650        | 0.1684        | 0.2118        | 0.2151        | 0.1981        | 0.1915        | 0.1874        | 0.1923        | 0.3477        | 0.1989        | 0.1854        | 0.2440        |
| dare.linear(Yu et al. 2024)             | 0.2832        | 0.2856        | 0.2392        | 0.2693        | 0.6357        | 0.4841        | 0.6288        | 0.5829        | 0.2018        | 0.2588        | 0.1650        | 0.2085        | 0.2545        | 0.2245        | 0.2537        | 0.2442        | 0.2162        | 0.1728        | 0.1608        | 0.1833        |
| dare.ties(Yu et al. 2024)               | 0.3194        | 0.1935        | 0.1780        | 0.2303        | 0.5375        | 0.4825        | 0.5558        | 0.5253        | 0.1851        | 0.1515        | 0.1859        | 0.1742        | 0.1790        | 0.1352        | 0.1730        | 0.1624        | 0.2932        | 0.1635        | 0.1890        | 0.2152        |
| regmean(Jin et al. 2025)                | 0.3460        | 0.2369        | 0.3407        | 0.3079        | 0.5812        | 0.2843        | 0.5022        | 0.4559        | 0.2405        | 0.1340        | 0.1388        | 0.1711        | 0.2504        | 0.1623        | 0.1493        | 0.1873        | 0.2276        | 0.1695        | 0.1318        | 0.1763        |
| fisher(Matena and Raffel 2022)          | 0.3196        | 0.2237        | 0.2608        | 0.2680        | 0.6707        | 0.6702        | 0.6690        | 0.6700        | 0.2973        | 0.1741        | 0.1812        | 0.2175        | 0.2010        | 0.2956        | 0.1963        | 0.2310        | 0.2333        | 0.1860        | 0.2513        | 0.2235        |
| breadcrumbs(Davari and Belilovsky 2024) | 0.0854        | 0.1417        | 0.1460        | 0.1244        | 0.0495        | 0.2913        | 0.1721        | 0.1710        | 0.0766        | 0.0676        | 0.0685        | 0.0709        | 0.0802        | 0.0948        | 0.0603        | 0.0784        | 0.1225        | 0.1273        | 0.1021        | 0.1173        |
| model.stock(Jang, Yun, and Han 2025)    | 0.2420        | 0.1824        | 0.1975        | 0.2073        | 0.6216        | 0.4311        | 0.5721        | 0.5416        | 0.1225        | 0.1274        | 0.1185        | 0.1228        | 0.2742        | 0.1578        | 0.2012        | 0.2111        | 0.1791        | 0.1321        | 0.1578        | 0.1563        |
| from(Ilharco et al. 2023)               | 0.2989        | 0.2306        | 0.2052        | 0.2449        | 0.6713        | 0.4369        | 0.6580        | 0.5887        | 0.2350        | 0.1560        | 0.1520        | 0.1810        | 0.2207        | 0.2087        | 0.2476        | 0.2257        | 0.2309        | 0.1506        | 0.2063        | 0.1959        |
| iso.c(Marczak et al. 2025)              | 0.3278        | 0.2748        | 0.3023        | 0.3016        | 0.5049        | 0.4422        | 0.6416        | 0.5296        | 0.2331        | 0.1629        | 0.1383        | 0.1781        | 0.2736        | 0.2118        | 0.2453        | 0.2436        | 0.2615        | 0.1698        | 0.1554        | 0.1956        |
| free.merge(Zheng and Wang 2025)         | 0.3345        | 0.2467        | 0.2822        | 0.2878        | 0.6291        | 0.4700        | 0.6246        | 0.5746        | 0.2261        | 0.1771        | 0.1396        | 0.1809        | 0.2863        | 0.1793        | 0.2475        | 0.2377        | 0.2435        | 0.1581        | 0.1545        | 0.1854        |
| robustmerge(Zeng et al. 2025)           | 0.3401        | 0.2430        | 0.2256        | 0.2696        | 0.2956        | 0.2085        | 0.3942        | 0.2994        | 0.2458        | 0.1764        | 0.1653        | 0.1958        | 0.2680        | 0.2227        | 0.2301        | 0.2403        | 0.2189        | 0.2231        | 0.1707        | 0.2042        |
| <b>LAMP-Merge</b>                       | <b>0.8002</b> | <b>0.7999</b> | <b>0.8007</b> | <b>0.8003</b> | <b>0.6745</b> | <b>0.6761</b> | <b>0.6763</b> | <b>0.6756</b> | <b>0.6074</b> | <b>0.6069</b> | <b>0.6076</b> | <b>0.6073</b> | <b>0.5531</b> | <b>0.5524</b> | <b>0.5517</b> | <b>0.5524</b> | <b>0.4762</b> | <b>0.4762</b> | <b>0.4762</b> | <b>0.4762</b> |

Table 2: Client-average test accuracy on ConvNeXt. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over  $K = 3, 5, 7$  within the corresponding dataset.

| Method                                  | Blood         |               |               |               | Derma         |               |               |               | Organ-C       |               |               |               | Organ-S       |               |               |               | Ultrasound    |               |               |               |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                                         | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           |
| avg(Wortsman et al. 2022)               | 0.1565        | 0.1122        | 0.1384        | 0.1357        | 0.2966        | 0.4830        | 0.6688        | 0.4828        | 0.1037        | 0.1218        | 0.0944        | 0.1066        | 0.1308        | 0.1035        | 0.1416        | 0.1253        | 0.1506        | 0.1506        | 0.1252        | 0.1421        |
| ties(Yadav et al. 2023)                 | 0.1715        | 0.1214        | 0.1524        | 0.1484        | 0.2966        | 0.1797        | 0.2713        | 0.2492        | 0.1217        | 0.0786        | 0.0716        | 0.0906        | 0.1123        | 0.0761        | 0.0796        | 0.0893        | 0.1518        | 0.1456        | 0.1291        | 0.1422        |
| dare.linear(Yu et al. 2024)             | 0.1756        | 0.1122        | 0.1345        | 0.1408        | 0.2705        | 0.4830        | 0.4830        | 0.4122        | 0.1035        | 0.0699        | 0.1085        | 0.0940        | 0.1282        | 0.1035        | 0.0797        | 0.1038        | 0.1575        | 0.1575        | 0.1264        | 0.1471        |
| dare.ties(Yu et al. 2024)               | 0.1096        | 0.1519        | 0.1497        | 0.1371        | 0.0908        | 0.2798        | 0.2451        | 0.2052        | 0.1305        | 0.1346        | 0.0770        | 0.1140        | 0.1096        | 0.0946        | 0.0709        | 0.0917        | 0.1456        | 0.1297        | 0.1500        | 0.1418        |
| regmean(Jin et al. 2025)                | 0.1565        | 0.0817        | 0.1856        | 0.1413        | 0.2633        | 0.0773        | 0.2720        | 0.2042        | 0.1143        | 0.1017        | 0.0768        | 0.0976        | 0.1014        | 0.1259        | 0.1038        | 0.1104        | 0.1575        | 0.1539        | 0.1500        | 0.1538        |
| fisher(Matena and Raffel 2022)          | 0.1565        | 0.0999        | 0.1189        | 0.1251        | 0.4825        | 0.2638        | 0.2971        | 0.3478        | 0.1003        | 0.0974        | 0.0915        | 0.0964        | 0.0604        | 0.0517        | 0.0843        | 0.0655        | 0.1695        | 0.1360        | 0.1456        | 0.1504        |
| breadcrumbs(Davari and Belilovsky 2024) | 0.1516        | 0.1366        | 0.1163        | 0.1348        | 0.0524        | 0.2766        | 0.4630        | 0.3647        | 0.1031        | 0.1444        | 0.1574        | 0.1350        | 0.1516        | 0.1554        | 0.2077        | 0.1716        | 0.1384        | 0.1363        | 0.1363        | 0.1370        |
| model.stock(Jang, Yun, and Han 2025)    | 0.1756        | 0.1144        | 0.1778        | 0.1559        | 0.2966        | 0.4830        | 0.6688        | 0.4828        | 0.1276        | 0.1218        | 0.1385        | 0.1293        | 0.1308        | 0.1035        | 0.1935        | 0.1426        | 0.1575        | 0.1291        | 0.1479        | 0.1448        |
| from(Ilharco et al. 2023)               | 0.1565        | 0.1083        | 0.1671        | 0.1440        | 0.2966        | 0.3406        | 0.4569        | 0.3647        | 0.1232        | 0.1224        | 0.0650        | 0.1035        | 0.1324        | 0.1105        | 0.0897        | 0.1109        | 0.1575        | 0.1578        | 0.1087        | 0.1413        |
| iso.c(Marczak et al. 2025)              | 0.1947        | 0.1865        | 0.1480        | 0.1764        | 0.2971        | 0.4830        | 0.6688        | 0.4830        | 0.0991        | 0.0699        | 0.0945        | 0.0878        | 0.0789        | 0.1477        | 0.0867        | 0.1044        | 0.1518        | 0.1291        | 0.1605        | 0.1471        |
| free.merge(Zheng and Wang 2025)         | 0.1575        | 0.1190        | 0.1384        | 0.1383        | 0.2971        | 0.4830        | 0.6688        | 0.4830        | 0.0906        | 0.1218        | 0.1385        | 0.1170        | 0.1308        | 0.1035        | 0.1416        | 0.1253        | 0.1506        | 0.1291        | 0.1695        | 0.1497        |
| robustmerge(Zeng et al. 2025)           | 0.1410        | 0.1405        | 0.0752        | 0.1189        | 0.0908        | 0.4507        | 0.4830        | 0.3415        | 0.1394        | 0.1218        | 0.1793        | 0.1468        | 0.2354        | 0.0961        | 0.2077        | 0.1797        | 0.1300        | 0.1135        | 0.1363        | 0.1266        |
| <b>LAMP-Merge</b>                       | <b>0.8482</b> | <b>0.8483</b> | <b>0.8493</b> | <b>0.8486</b> | <b>0.5992</b> | <b>0.5920</b> | <b>0.5852</b> | <b>0.5921</b> | <b>0.6482</b> | <b>0.6480</b> | <b>0.6495</b> | <b>0.6486</b> | <b>0.5977</b> | <b>0.5985</b> | <b>0.5990</b> | <b>0.5984</b> | <b>0.4259</b> | <b>0.4259</b> | <b>0.4259</b> | <b>0.4259</b> |

Table 3: Client-average test accuracy on ViT-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over  $K = 3, 5, 7$  within the corresponding dataset.

| Method                                  | Blood         |               |               |               | Derma         |               |               |               | Organ-C       |               |               |               | Organ-S       |               |               |               | Ultrasound    |               |               |               |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                                         | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           | $K = 3$       | $K = 5$       | $K = 7$       | Avg           |
| avg(Wortsman et al. 2022)               | 0.1343        | 0.1082        | 0.1160        | 0.1195        | 0.4638        | 0.4369        | 0.3696        | 0.4234        | 0.1657        | 0.1043        | 0.0952        | 0.1217        | 0.0844        | 0.0852        | 0.1710        | 0.1135        | 0.1450        | 0.1327        | 0.1884        | 0.1554        |
| ties(Yadav et al. 2023)                 | 0.0796        | 0.1421        | 0.1285        | 0.1167        | 0.4780        | 0.3498        | 0.4615        | 0.4298        | 0.1341        | 0.1223        | 0.0953        | 0.1172        | 0.1192        | 0.1002        | 0.1597        | 0.1264        | 0.2001        | 0.1306        | 0.1018        | 0.1442        |
| dare.linear(Yu et al. 2024)             | 0.2397        | 0.1129        | 0.1387        | 0.1638        | 0.2850        | 0.1002        | 0.1204        | 0.1685        | 0.1534        | 0.0946        | 0.0704        | 0.1061        | 0.0748        | 0.0549        | 0.1611        | 0.0969        | 0.1366        | 0.1051        | 0.1435        | 0.1284        |
| dare.ties(Yu et al. 2024)               | 0.0735        | 0.1189        | 0.1146        | 0.1023        | 0.2735        | 0.4502        | 0.3528        | 0.3588        | 0.1252        | 0.0903        | 0.1033        | 0.1063        | 0.0541        | 0.0874        | 0.0591        | 0.0669        | 0.1447        | 0.1120        | 0.1042        | 0.1203        |
| regmean(Jin et al. 2025)                | 0.1748        | 0.1417        | 0.1290        | 0.1485        | 0.5114        | 0.3667        | 0.6658        | 0.5146        | 0.1250        | 0.0964        | 0.0983        | 0.1066        | 0.1263        | 0.0717        | 0.1068        | 0.1016        | 0.1491        | 0.1384        | 0.1144        | 0.1340        |
| fisher(Matena and Raffel 2022)          | 0.1166        | 0.1512        | 0.1944        | 0.1541        | 0.4780        | 0.2042        | 0.4820        | 0.3881        | 0.0799        | 0.1072        | 0.1424        | 0.1098        | 0.0856        | 0.1218        | 0.1222        | 0.1099        | 0.1611        | 0.1315        | 0.1420        | 0.1449        |
| breadcrumbs(Davari and Belilovsky 2024) | 0.1363        | 0.1039        | 0.1300        | 0.1234        | 0.0489        | 0.2572        | 0.0607        | 0.1223        | 0.0773        | 0.1000        | 0.1644        | 0.1139        | 0.1237        | 0.0483        | 0.1219        | 0.0980        | 0.1360        | 0.1141        | 0.1524        | 0.1342        |
| model.stock(Jang, Yun, and Han 2025)    | 0.1940        | 0.1120        | 0.1134        | 0.1398        | 0.3805        | 0.3079        | 0.2735        | 0.3206        | 0.1488        | 0.1227        | 0.1215        | 0.1310        | 0.0893        | 0.0859        | 0.1867        | 0.1206        | 0.1599        | 0.1809        | 0.1641        | 0.1683        |
| from(Ilharco et al. 2023)               | 0.1802        | 0.0781        | 0.1405        | 0.1329        | 0.4695        | 0.5121        | 0.6688        | 0.5501        | 0.2148        | 0.1301        | 0.0972        | 0.1474        | 0.0772        | 0.0798        | 0.1485        | 0.1018        | 0.1836        | 0.1743        | 0.1776        | 0.1785        |
| iso.c(Marczak et al. 2025)              | 0.1234        | 0.1483        | 0.1018        | 0.1245        | 0.5709        | 0.4529        | 0.2306        | 0.4181        | 0.1084        | 0.0567        | 0.0897        | 0.0849        | 0.0611        | 0.0531        | 0.1081        | 0.0741        | 0.1417        | 0.1527        | 0.1869        | 0.1604        |
| free.merge(Zheng and Wang 2025)         | 0.1989        | 0.1457        | 0.1392        | 0.1613        | 0.4630        | 0.4770        | 0.3855        | 0.4418        | 0.1425        | 0.0980        | 0.0759        | 0.1055        | 0.0727        | 0.0914        | 0.1643        | 0.1095        | 0.1405        | 0.1593        | 0.1647        | 0.1548        |
| robustmerge(Zeng et al. 2025)           | 0.1482        | 0.1070        | 0.1425        | 0.1326        | 0.0347        | 0.2768        | 0.1724        | 0.1613        | 0.0954        | 0.0617        | 0.1438        | 0.1003        | 0.1260        | 0.0707        | 0.1217        | 0.1061        | 0.1698        | 0.1854        | 0.1563        | 0.1705        |
| <b>LAMP-Merge</b>                       | <b>0.8523</b> | <b>0.8522</b> | <b>0.8516</b> | <b>0.8520</b> | <b>0.5970</b> | <b>0.5938</b> | <b>0.5860</b> | <b>0.5923</b> | <b>0.5805</b> | <b>0.5790</b> | <b>0.5790</b> | <b>0.5795</b> | <b>0.5406</b> | <b>0.5372</b> | <b>0.5388</b> | <b>0.5389</b> | <b>0.4645</b> | <b>0.4645</b> | <b>0.4645</b> | <b>0.4645</b> |

Table 4: Client-average test accuracy on Swin-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over  $K = 3, 5, 7$  within the corresponding dataset.

| Method                                  | Blood  |        |        |        | Derma  |        |        |        | Organ-C |        |        |        | Organ-S |        |        |        | Ultrasound |        |        |        |
|-----------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|---------|--------|--------|--------|------------|--------|--------|--------|
|                                         | K = 3  | K = 5  | K = 7  | Avg    | K = 3  | K = 5  | K = 7  | Avg    | K = 3   | K = 5  | K = 7  | Avg    | K = 3   | K = 5  | K = 7  | Avg    | K = 3      | K = 5  | K = 7  | Avg    |
| avg(Wortsman et al. 2022)               | 0.1565 | 0.1186 | 0.1524 | 0.1425 | 0.4569 | 0.4830 | 0.6688 | 0.5362 | 0.1068  | 0.1294 | 0.0787 | 0.1050 | 0.0682  | 0.0604 | 0.1381 | 0.0889 | 0.1527     | 0.1294 | 0.1465 | 0.1429 |
| ties(Yadav et al. 2023)                 | 0.0972 | 0.1190 | 0.1384 | 0.1182 | 0.0841 | 0.0780 | 0.6688 | 0.2770 | 0.1160  | 0.0702 | 0.0934 | 0.0932 | 0.0734  | 0.0607 | 0.1107 | 0.0816 | 0.1695     | 0.1303 | 0.1261 | 0.1420 |
| dare_linear(Yu et al. 2024)             | 0.1565 | 0.1161 | 0.1825 | 0.1517 | 0.4569 | 0.2771 | 0.4630 | 0.3990 | 0.1155  | 0.1272 | 0.0776 | 0.1068 | 0.0640  | 0.0720 | 0.1381 | 0.0914 | 0.1743     | 0.1536 | 0.1497 | 0.1592 |
| dare_ties(Yu et al. 2024)               | 0.1384 | 0.1162 | 0.1384 | 0.1310 | 0.2961 | 0.2638 | 0.6688 | 0.4096 | 0.1276  | 0.0744 | 0.1116 | 0.1055 | 0.1194  | 0.0434 | 0.1230 | 0.0953 | 0.1791     | 0.1351 | 0.1300 | 0.1481 |
| regmean(Jin et al. 2025)                | 0.1374 | 0.1715 | 0.1578 | 0.1556 | 0.2705 | 0.4630 | 0.4569 | 0.3968 | 0.1411  | 0.0773 | 0.0811 | 0.0998 | 0.1215  | 0.0821 | 0.0926 | 0.0987 | 0.1465     | 0.1105 | 0.1779 | 0.1450 |
| fisher(Matena and Raffel 2022)          | 0.1633 | 0.1756 | 0.1565 | 0.1651 | 0.4582 | 0.0652 | 0.4718 | 0.3317 | 0.0814  | 0.0899 | 0.0978 | 0.0897 | 0.0651  | 0.0650 | 0.1092 | 0.0798 | 0.1746     | 0.1165 | 0.1755 | 0.1555 |
| breadcrumbs(Davari and Belilovsky 2024) | 0.1500 | 0.1365 | 0.1537 | 0.1467 | 0.0647 | 0.2771 | 0.0115 | 0.1378 | 0.1445  | 0.1783 | 0.1660 | 0.1629 | 0.2354  | 0.2077 | 0.2111 | 0.2181 | 0.1450     | 0.1312 | 0.1653 | 0.1472 |
| model_stock(Jang, Yun, and Han 2025)    | 0.1374 | 0.1440 | 0.1715 | 0.1510 | 0.4569 | 0.2771 | 0.4825 | 0.4055 | 0.1110  | 0.0771 | 0.1404 | 0.1095 | 0.0671  | 0.1106 | 0.1714 | 0.1164 | 0.1539     | 0.1273 | 0.1869 | 0.1560 |
| from(Illharco et al. 2023)              | 0.1565 | 0.1495 | 0.1671 | 0.1577 | 0.4569 | 0.2971 | 0.6688 | 0.4743 | 0.1174  | 0.1220 | 0.1051 | 0.1148 | 0.1133  | 0.1016 | 0.1380 | 0.1176 | 0.1539     | 0.1252 | 0.1476 | 0.1422 |
| iso_c(Marczak et al. 2025)              | 0.1430 | 0.1722 | 0.1700 | 0.1617 | 0.3099 | 0.2840 | 0.5174 | 0.3704 | 0.1302  | 0.1327 | 0.1469 | 0.1366 | 0.1545  | 0.1727 | 0.1880 | 0.1717 | 0.1755     | 0.1731 | 0.1791 | 0.1759 |
| free_merge(Zheng and Wang 2025)         | 0.1154 | 0.0861 | 0.1840 | 0.1285 | 0.4569 | 0.4830 | 0.6688 | 0.5362 | 0.0959  | 0.1377 | 0.0890 | 0.1075 | 0.0682  | 0.0497 | 0.1062 | 0.0747 | 0.1518     | 0.1575 | 0.1599 | 0.1564 |
| robustmerge(Zeng et al. 2025)           | 0.1374 | 0.1409 | 0.1670 | 0.1484 | 0.2377 | 0.2771 | 0.0504 | 0.1884 | 0.1601  | 0.1442 | 0.1960 | 0.1668 | 0.1673  | 0.1151 | 0.2077 | 0.1634 | 0.1761     | 0.1315 | 0.1629 | 0.1568 |
| LAMP-Merge                              | 0.0737 | 0.7753 | 0.7748 | 0.7746 | 0.6374 | 0.6251 | 0.6244 | 0.6290 | 0.6651  | 0.6632 | 0.6630 | 0.6638 | 0.6078  | 0.6076 | 0.6068 | 0.6074 | 0.4816     | 0.4816 | 0.4816 | 0.4816 |



Table 5: Internal ablation of diagnostic prototype reconstruction. All settings cover 180 raw cells and 60 client-average cells.

| Setting                     | Mean Acc      | $\Delta$ vs. LAMP   |
|-----------------------------|---------------|---------------------|
| <b>LAMP-Merge</b>           | <b>0.6209</b> | <b>—</b>            |
| Classifier-head aggregation | 0.2579        | $\downarrow 0.3630$ |
| Global-feature mean         | 0.2645        | $\downarrow 0.3564$ |
| Support-only synthetic head | 0.1137        | $\downarrow 0.5073$ |
| Shuffled-label prototype    | 0.2002        | $\downarrow 0.4207$ |
| Uniform client weight       | 0.5839        | $\downarrow 0.0370$ |
| Binary support only         | 0.5517        | $\downarrow 0.0692$ |
| Global client-size weight   | 0.5944        | $\downarrow 0.0266$ |

Table 6: Internal ablation of long-tail prevalence calibration.

| Setting                   | Mean Acc      | $\Delta$ vs. LAMP   |
|---------------------------|---------------|---------------------|
| <b>LAMP-Merge</b>         | <b>0.6209</b> | <b>—</b>            |
| No prevalence calibration | 0.5879        | $\downarrow 0.0330$ |
| Uniform prevalence prior  | 0.5879        | $\downarrow 0.0330$ |
| Smoothed prevalence prior | 0.6209        | 0.0000              |

## Ablation Study

Module-level ablations use exactly the same experimental setup as the main experiments. We compare three configurations: full LAMP-Merge, diagnostic prototype reconstruction only, and weight averaging augmented only with long-tail prevalence calibration. Both components provide substantial performance improvements.

### 1) Effectiveness of Diagnostic Prototype Reconstruction

To further identify which information within diagnostic prototype reconstruction is necessary, we conduct controlled substitutions across the full setting. All configurations retain long-tail prevalence calibration and vary only the prototype semantics or support-count weighting.

Fig. 8 and Table 5 show that neither an arbitrary head nor a class-agnostic statistic explains the gain. Replacing reference-space prototypes or class-specific support weights consistently reduces ACC, confirming that **diagnostic semantics and class-level reliability are jointly necessary**.

### 2) Effectiveness of Long-tail Prevalence Calibration

We further replace the prior term inside long-tail prevalence calibration. This group changes only the class prior used by the calibration bias.

Table 6 shows that removing prevalence calibration or replacing the real prior with a uniform prior reduces mean ACC, confirming that long-tailed prevalence is part of the full method. The smoothed prior remains close to the official method, indicating that LAMP-Merge does not depend on accidental fluctuations of exact counts; bounded calibration remains effective as long as the prior preserves the real long-tailed structure.

## Conclusion

We studied **one-shot asynchronous medical model merging**, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical anal-

ysis shows that generic model-merging methods suffer from **aggregation-induced predictive collapse** under incomplete local diagnostic coverage. We further show that, on long-tailed medical datasets, such collapse can be masked by raw accuracy. We therefore introduce the Long-tail Prevalence Calibration module and perform secondary evaluation using macro recall, macro-F1, and predictive-distribution diagnostics (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

Taken together, the full-scope comparisons, prediction-distribution diagnostics, and prototype-error analysis establish a consistent mechanism: class-level support-weighted evidence reconstructs discriminative directions for locally absent diagnoses, while bounded prevalence calibration retains clinically meaningful imbalance without reintroducing majority-class collapse.

LAMP-Merge addresses these two properties through a concise two-module design. Future work will extend its applicability in two directions: first, from medical image classification to segmentation, detection, and multimodal diagnostic tasks; and second, toward stronger privacy-preserving mechanisms, such as differential privacy and secure aggregation, to further reduce potential privacy risks associated with class-level statistics (Kaissis et al. 2020a; Geyer, Klein, and Nabi 2017; Abadi et al. 2016). We will also explore deployment and validation in larger-scale real-world multicenter healthcare settings.

Table 7: Prediction-collapse diagnostics averaged over five medical datasets and four backbones per dataset.

| Metric                       | LAMP-Merge    | Best reference | $\Delta$ vs. LAMP   |
|------------------------------|---------------|----------------|---------------------|
| Accuracy $\uparrow$          | <b>0.5885</b> | 0.3156         | $\uparrow+0.2729$   |
| BA / macro recall $\uparrow$ | <b>0.5747</b> | 0.1810         | $\uparrow+0.3937$   |
| Macro F1 $\uparrow$          | <b>0.5352</b> | 0.1123         | $\uparrow+0.4229$   |
| Collapse ratio $\downarrow$  | <b>0.2294</b> | 0.7477         | $\downarrow-0.5184$ |
| Effective classes $\uparrow$ | <b>8.0390</b> | 2.1360         | $\uparrow+5.9030$   |
| Pred-true TV $\downarrow$    | <b>0.1510</b> | 0.6168         | $\downarrow-0.4658$ |

## Supplementary Results and Analysis

### Hyperparameter Analysis Experiment

Fig. 3 evaluates the prototype-head scale  $s$  and calibration strength  $\lambda$  over 180 raw cells and 60 client-average cells, spanning five datasets, four vision backbones, three client counts, and three Dirichlet skew levels. The default setting is  $s = 20$  and  $\lambda = 5$ .

Both curves remain stable over broad intervals. Small  $s$  is insufficient to scale prototype logits, whereas bounded  $\lambda$  calibrates prevalence without inducing majority-class prediction. Thus, the gain arises from the two components rather than narrow hyperparameter tuning.

### New-metric Diagnostic Experiment

Although the main experiments use ACC for direct comparison, long-tailed medical distributions can allow single-class or few-class predictors to achieve high ACC (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). We therefore report balanced accuracy, macro-F1, collapse ratio, effective predicted classes, and predicted-true total variation.

Table 7 compares LAMP-Merge with the best-performing baseline for each metric. **LAMP-Merge consistently and substantially outperforms the existing methods.** This indicates that its accuracy gains do not arise from more severe single-class prediction collapse, but rather from more complete global diagnostic discrimination.

### Visualization Experiment

To visualize the prediction-distribution changes measured by the new diagnostic metrics, we compare the predicted class allocation induced by LAMP-Merge with that of the strongest generic merge. The generic merge concentrates predictions on one class, whereas LAMP-Merge recovers a broader class allocation aligned with the test distribution.

### t-SNE Visualization Experiment

We use t-SNE (van der Maaten and Hinton 2008) only for analysis, not for server-side selection or fusion. The joint comparison uses Blood with ResNet,  $K = 3$ , and  $\beta = 0.01$ . Each point is the output-probability representation of a test image, and formal merging methods are projected into the same two-dimensional space to compare the class structure of their predictive representations. LAMP-Merge retains separated predictive groups, whereas generic merges exhibit a substantially compressed output space.

## In-depth Analysis Experiment

**1) analysis of the geometry** Finally, we conduct an in-depth analysis of the geometry induced by diagnostic prototype reconstruction. This experiment corresponds to the prototype-replacement settings in the internal ablation and explains why reference-space class prototypes and class-level support counts jointly provide stable gains. We assess inter-class separation, nearest-class margin, within-class client consistency, and prototype-client alignment.

Across the controlled prototype substitutions, the full method jointly preserves nonzero inter-class separation and high cross-client consistency and prototype-client alignment, rather than maximizing a single geometric value. Together with the accuracy drops in Table 5, this joint profile shows that the benefit arises from class-semantic reference-space prototypes weighted by class-specific support.

The individual geometry values should therefore be interpreted as mechanism diagnostics, not standalone objectives: extreme separation or perfect alignment can be produced by class-agnostic or mismatched directions. LAMP-Merge is effective because its four quantities are simultaneously consistent with class semantics and the observed full-scope accuracy gains.

**2) Additional Theoretical Analysis** Let  $\mu_c^*$  denote the true class mean in the shared reference space. If client class features have bounded second moments, the prototype estimation error satisfies

$$\mathbb{E}\|p_c - \mu_c^*\|_2^2 \leq \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \text{Bias}_c^2. \quad (10)$$

This upper bound shows that support counts directly control the variance of class-prototype estimation. A client that does not observe class  $c$  should not contribute evidence for that class, and a client with more samples should contribute more, but only sublinearly.

Let the true margin between classes  $c$  and  $d$  for sample  $x$  be

$$\Delta_{c,d}(x) = (\mu_c^*)^\top z - (\mu_d^*)^\top z. \quad (11)$$

When the prototype error satisfies

$$\|p_c - \mu_c^*\|_2 + \|p_d - \mu_d^*\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \quad (12)$$

the estimated prototypes preserve the class order between  $c$  and  $d$ .

The role of long-tail prevalence calibration is to introduce long-tail priors in a bounded manner. For prevalence counts  $m_{i,c}$ , the class prior is

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \quad (13)$$

and the centered log-prior bias is

$$b_c = \lambda \left( \log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \quad (14)$$

As long as  $\lambda$  is bounded, long-tail prevalence calibration does not replace the discriminative directions reconstructed

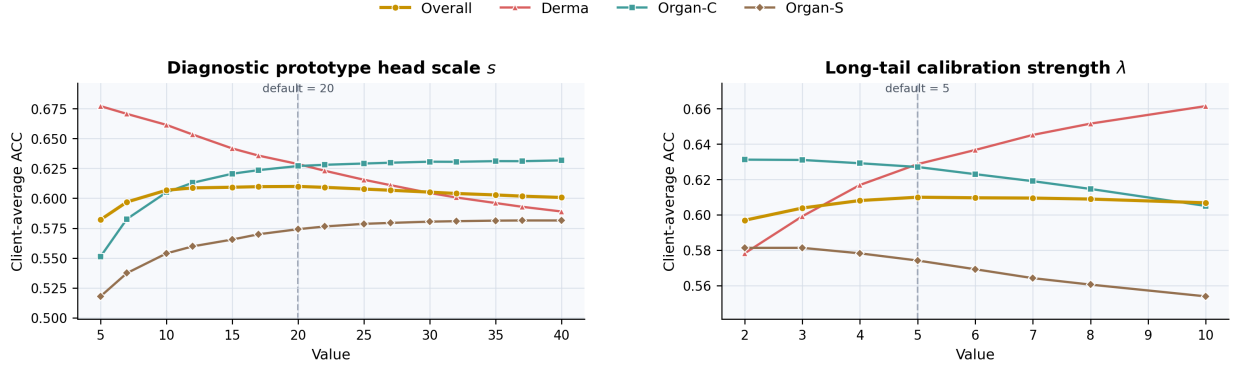


Figure 3: Full-scope hyperparameter sensitivity. The default setting  $s = 20$  and  $\lambda = 5$  lies in the stable high-accuracy plateau.

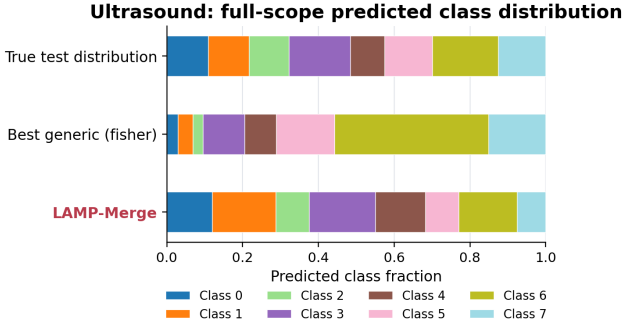


Figure 4: Predicted class distributions on Ultrasound, averaged over the four visual backbones,  $K \in \{3, 5, 7\}$ , and the three Dirichlet skew levels. The generic merge concentrates predictions on one class, while LAMP-Merge recovers a broader class allocation aligned with the test distribution.

by diagnostic prototype reconstruction; it only calibrates the output toward the real class prevalence. This explains why using long-tail prevalence calibration alone is insufficient, while full LAMP-Merge remains both non-collapsed and prevalence-aware.

## Evaluation Datasets

The five MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023) cover blood-cell microscopy, dermoscopy, abdominal CT organ slices, and ultrasound images. They involve cell morphology recognition, skin-lesion diagnosis, organ-structure recognition, and ultrasound-structure discrimination:

- **Blood:** an eight-class blood-cell microscopy dataset for hematological morphology classification, focusing on fine-grained cellular appearance and inter-class morphological similarity.
- **Derma:** a seven-class dermoscopic skin-lesion dataset, characterized by strong long-tailed prevalence and visually subtle lesion categories.
- **Organ-C:** an 11-class abdominal CT dataset in the coronal view, evaluating organ-category recognition under

grayscale cross-sectional imaging.

- **Organ-S:** an 11-class abdominal CT dataset in the sagittal view, complementing Organ-C with a different anatomical imaging plane.
- **Ultrasound:** an eight-class ultrasound image dataset, covering structure recognition under modality-specific speckle noise and low-contrast boundaries.

Together, these benchmarks provide a systematic testbed for evaluating cross-modality robustness, class-discriminative capability, and task generalization under different medical image types and class-distribution conditions.

## Detailed Ablation Configurations

All diagnostic-prototype substitutions retain long-tail prevalence calibration and vary only prototype semantics or support-count weighting:

- **LAMP-Merge** uses  $p_c = \sum_i \alpha_{i,c} \mu_{i,c}$ , where  $e_{i,c} = (n_{i,c} + 1)^{\gamma} \mathbf{1}[n_{i,c} > 0]$  determines the class-support reliability.
- **Classifier-head aggregation** replaces the reference-space class prototype by the local classifier-head direction, i.e.,  $\mu_{i,c} \leftarrow h_{i,c}$ .
- **Global-feature mean** removes class conditioning by setting  $\mu_{i,c} \leftarrow g$  for all  $c$ , where  $g$  is the class-agnostic global reference-feature mean.
- **Support-only synthetic head** removes image evidence by replacing each prototype with a fixed random unit direction  $u_c$ , i.e.,  $p_c \leftarrow u_c, \|u_c\|_2 = 1$ .
- **Shuffled-label prototype** preserves prototype magnitudes but permutes diagnostic semantics, i.e.,  $p_c \leftarrow p_{\sigma(c)}$ , where  $\sigma$  is a random permutation over diagnosis labels.
- **Uniform client weight** replaces support-weighted aggregation with equal weights among clients containing class  $c$ , i.e.,  $\alpha_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0] / \sum_j \mathbf{1}[n_{j,c} > 0]$ .
- **Binary support only** replaces fine-grained support and prevalence counts with class-presence indicators, i.e.,  $n_{i,c}, m_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0]$ .



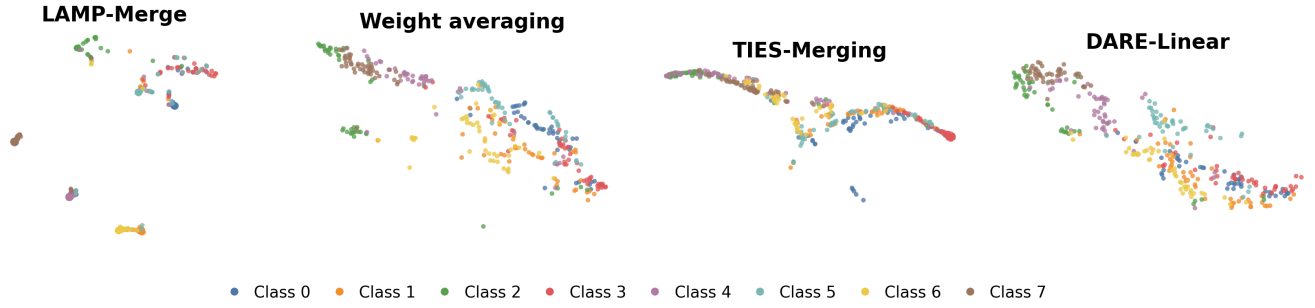


Figure 5: Joint t-SNE visualization of output-probability representations on Blood. From left to right: LAMP-Merge, weight averaging, TIES-Merging, and DARE-Linear.

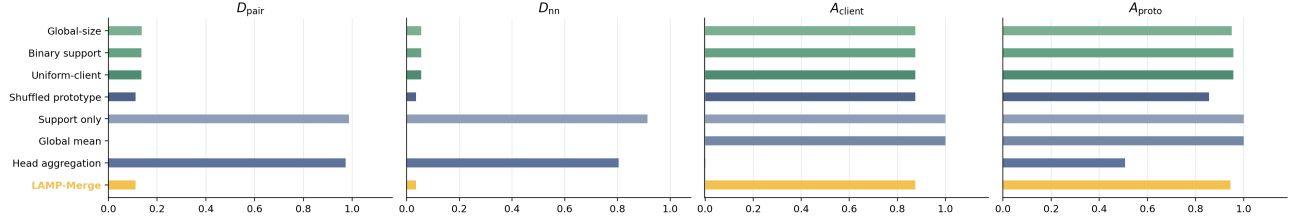


Figure 6: Prototype-geometry comparison on the full benchmark. The figure summarizes pairwise distance, nearest-class distance, prototype consistency, and prototype-client alignment.

- **Global client-size weight** uses the total client support  $N_i = \sum_k n_{i,k}$  rather than class-specific support, i.e.,  $e_{i,c} \leftarrow (N_i + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$ .

The long-tail prevalence substitutions vary only the class prior used by the calibration bias:

- **LAMP-Merge** estimates the empirical class prior as  $\pi_c = \sum_i m_{i,c} / \sum_k \sum_i m_{i,k}$ .
- **No prevalence calibration** removes the calibration bias by setting  $b_c \leftarrow 0$ .
- **Uniform prevalence prior** removes long-tail prevalence by setting  $\pi_c \leftarrow 1/C$ , where  $C$  is the number of diagnosis classes.
- **Smoothed prevalence prior** uses  $\pi_c \leftarrow (\sum_i (m_{i,c} + 1) / \sum_k \sum_i (m_{i,k} + 1))$ , preserving the same prior structure with additive smoothing.

### Metric Definitions and Geometry Details

Let  $y_n$  and  $\hat{y}_n$  denote the true and predicted labels of the  $n$ -th test sample. Let  $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$  be the predicted-class distribution,  $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$  the true class distribution, and  $\mathcal{C}_{\text{test}} = \{c \mid \sum_n \mathbf{1}[y_n = c] > 0\}$  the set of classes observed in the test set. We use:

- $\text{BA} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{Recall}_c$ , balanced accuracy or macro recall, which measures whether all observed diagnosis classes are recalled.
- $\text{MacroF1} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \text{F1}_c$ , class-balanced precision-recall quality that penalizes majority-only predictions.
- $\rho = \max_c q_c$ , the prediction-collapse ratio, which measures concentration on a single diagnosis class.

- $C_{\text{eff}} = \exp(-\sum_c q_c \log q_c)$ , the effective number of predicted classes.
- $\text{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|$ , the total variation distance between predicted and true class distributions.

Let  $d(a, b) = 1 - \cos(a, b)$  denote cosine distance and let  $\mathcal{I}_c = \{i \mid n_{i,c} > 0\}$  denote the clients that contain diagnosis class  $c$ . The geometric quantities are:

- $D_{\text{pair}} = \frac{2}{C(C-1)} \sum_{c < d} d(p_c, p_d)$ , the mean distance between global prototypes of different diagnosis classes.
- $D_{\text{nn}} = \frac{1}{C} \sum_c \min_{d \neq c} d(p_c, p_d)$ , the mean nearest-class distance and most vulnerable decision margin.
- $A_{\text{client}} = \frac{1}{C} \sum_c \text{Avg}_{i < j, i, j \in \mathcal{I}_c} \cos(\mu_{i,c}, \mu_{j,c})$ , the within-class consistency among client prototypes;
- $A_{\text{proto}} = \frac{1}{C} \sum_c \sum_{i \in \mathcal{I}_c} \alpha_{i,c} \cos(p_c, \mu_{i,c})$ , the alignment between the global prototype and contributing client prototypes.

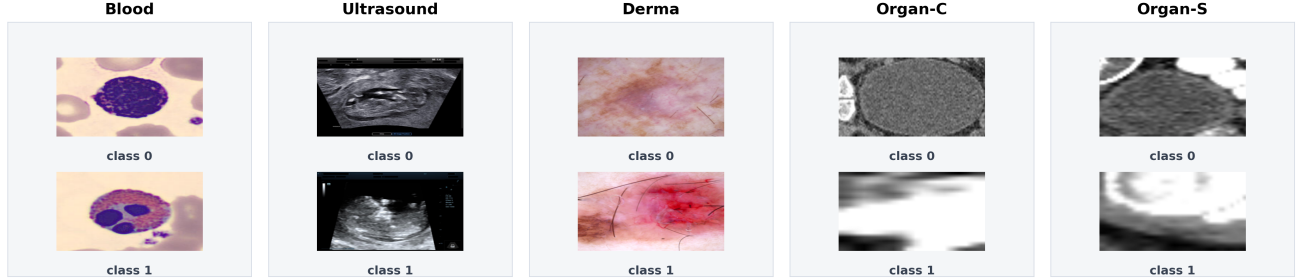


Figure 7: Representative training images from the five medical image datasets used in the experiments. The class IDs are shown only to illustrate category diversity within each dataset.

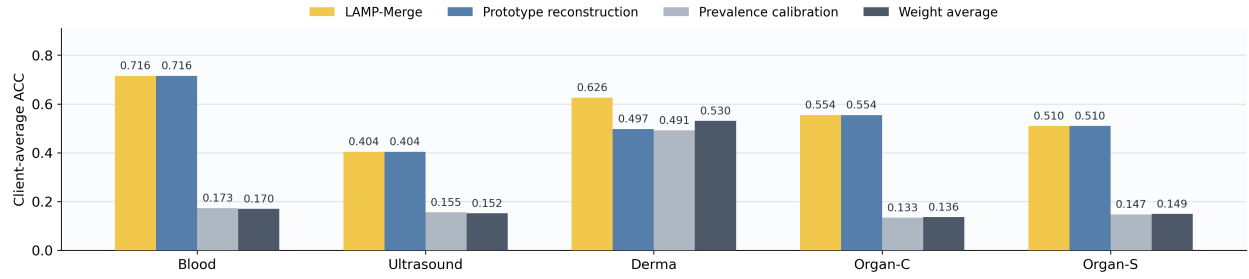


Figure 8: Dataset-level client-average accuracy in the module ablation. Each dataset contains 12 cells from four vision-model families and three client counts; each cell is averaged over three Dirichlet skew levels.

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